

Practical week 3

Session 3:

Task 1:

```
jella_pragathi@LAPTOP-RQQC: ~$ nano fork_process.c
GNU nano 8.7.1 fork_process.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n----- CHILD PROCESS ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(5);

        printf("Child process terminating.\n");
    }
    else
    {
        printf("\n----- PARENT PROCESS ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child completed.\n");
        printf("Parent process terminating.\n");
    }
}
```

```
else
{
    printf("\n----- PARENT PROCESS ----- \n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);

    wait(NULL);

    printf("Child completed.\n");
    printf("Parent process terminating.\n");
}

return 0;
}
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ nano fork_process.c
jella_pragathi@LAPTOP-RQQGT555:~$ gcc fork_process.c -o fork_process
jella_pragathi@LAPTOP-RQQGT555:~$ ./fork_process

----- PARENT PROCESS -----
Parent PID : 1027
Child PID  : 1028

----- CHILD PROCESS -----
Child PID  : 1028
Parent PID : 1027
Child is running...
Child process terminating.
Child completed.
Parent process terminating.
jella_pragathi@LAPTOP-RQQGT555:~$ |
```

Task 2:

```
jella_pragathi@LAPTOP-RQQGT555: ~$ nano state.c
GNU nano 8.7.1 state.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("PID = %d\n", getpid());

    sleep(120);    // wait for 120 seconds

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
jella_pragathi@LAPTOP-RQQGT555:~$ nano state.c
jella_pragathi@LAPTOP-RQQGT555:~$ gcc state.c -o state
jella_pragathi@LAPTOP-RQQGT555:~$ ./state &
[1] 1293
PID = 1293
jella_pragathi@LAPTOP-RQQGT555:~$ ^C
jella_pragathi@LAPTOP-RQQGT555:~$ ps -o pid,ppid,stat,cmd -p 1293
  PID   PPID  STAT  CMD
  1293   952   S     ./state
jella_pragathi@LAPTOP-RQQGT555:~$ cat /proc/1293/status | grep State
State:  S (sleeping)
jella_pragathi@LAPTOP-RQQGT555:~$ |
```